

A Machine-Checked Level-Four Bound on the Matrix Multiplication Exponent: A Tie-Manifold Search Case Study

BSD (bsd.developer@proton.me)
Independent Researcher

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Abstract

We give an exact, machine-checked certificate for the level-four bound $\omega_{\mathbb{Q}} \leq 43740440563687076465/2^{64} = 2.37117403423113 \dots$, below the published bound 2.371177. The numerical search and certification ran on a single Apple M1 Max laptop with 64 GiB unified memory, without GPU or cloud compute; the search used CPU float64. Lean 4 checks the fixed certificate, and an independent exact Rust checker agrees. The inference from verified feasibility to the exponent bound uses Alman et al.’s published combination-loss theorem, cited but not formalized in the released Lean package. We present the search as a case study: at the final numerical point, twenty three-way minima nearly tie, and measured branch-contrast gradients and fitted convex weights motivate reduced tangent models with finite balance corrections. An elementary branch-mixture identity, proved in Lean, separates surrogate gain from the slack paid when a move leaves the tie manifold. Proof-based tests also confirmed silently missing curvature in automatic differentiation. The native-evaluation dependencies and remaining Lean axioms are disclosed explicitly.

1. Introduction and result

Upper bounds on the matrix multiplication exponent ω have come, for four decades, from the laser method (Coppersmith and Winograd 1990; Stothers 2010; Vassilevska Williams 2012; Le Gall 2014; Alman and Vassilevska Williams 2021). Combination loss analysis (Duan, Wu, and Zhou 2023; Vassilevska Williams et al. 2024; Alman et al. 2025) reduces the bound to a finite optimization problem: any feasible point of its stated program yields $\omega \leq \Omega$ (Alman et al. 2025). Alman et al. obtained $\omega < 2.371339$ at level three. Dupont et al. (Dupont et al. 2026) solved the level-four instance with a tensor implementation and gradient-based search, refined by AlphaEvolve (Novikov et al. 2025), and reported $\omega < 2.371177$.

We start from Dupont et al.’s level-four formulation of Alman et al.’s combination-loss optimization problem (Dupont et al. 2026; Alman et al. 2025). Our numerical search and exact certification ran on a single Apple M1 Max laptop, without GPU or cloud compute. At the same level ($q = 5, \ell^* = 4$), our exact certificate and Lean theorem give

$$\omega \leq \frac{43740440563687076465}{2^{64}} = 2.37117403423113052 \dots < 2.371177.$$

Our certified bound is approximately 2.966×10^{-6} below the published bound of (Dupont et al. 2026). Section 6 states the theorem’s assumptions and Appendix C explains how to verify the selected certificate package. Throughout this paper, ω denotes the rational-field matrix multiplication exponent $\omega_{\mathbb{Q}}$ used in the formal theorem.

The methodological contribution is a case study of an adaptive search trajectory and its observed behavior. Near the best points, twenty minima over the tensor factors X, Y, Z nearly tie. A smooth weighted-branch surrogate can improve while the original objective worsens, because a move pays branch slack when it breaks those ties. We measured this geometry, used exact identities and conditional Lean lemmas to sharpen the experimental questions, and searched with reduced second-order models along the approximate tie manifold. The finite endpoints were always ranked by the original hard score.

2. The level-four objective

At level $\ell^* = 4$ with $q = 5$, a feasible point assigns distributions and scalar parameters to a recursion tree. Paths remain distinct even when their shapes and regions coincide; Appendix A summarizes the construction. Dupont et al. (Dupont et al. 2026) use Sinkhorn maximum-entropy solves (Sinkhorn and Knopp 1967; Cuturi 2013), implicit differentiation (Eisenberger et al. 2022), and Adam (Kingma and Ba 2015) in JAX (Bradbury et al. 2018) over roughly seven million GPU parameters. Our search used PyTorch (Paszke et al. 2019) in CPU float64 with a native maximum-entropy solver.

Let $p \in \mathcal{P}$ collect the free distributions and leaf parameters, where $\mathcal{P}$ is a product of simplices and intervals. The numerical search uses $n = 4,664,058$ active raw coordinates $\theta \in \mathbb{R}^n$; softmax and sigmoid maps, with inactive coordinates fixed, produce the physical point $p = \Phi(\theta) \in \mathcal{P}$. We abbreviate $E(\Phi(\theta))$ and $M(\Phi(\theta))$ as $E(\theta)$ and $M(\theta)$. The exact certificate concerns the physical point p . For a target τ , the *surplus* is

$$F_\tau(\theta) = E(\theta) + \tau M(\theta) - C, \quad C = 2^{\ell^*-1} \log_2(q+2) = 8 \log_2 7. \quad (1)$$

A point with $F_\tau(\theta) \geq 0$ is feasible for τ , and by the theorem of (Alman et al. 2025) gives $\omega \leq \tau$. When $M(\theta) > 0$, the smallest target a point supports is

$$\Omega_{\text{hard}}(\theta) = \max\left\{0, \frac{C - E(\theta)}{M(\theta)}\right\}. \quad (2)$$

Equation (2) is the numerical score of the search. It is computed in floating point and is not itself a proof. §6 describes the exact check that replaces it.

The objective’s nonsmoothness comes from precisely twenty three-way minima: six root sites, twelve interior sites, a level-two exponent site, and a matrix-size site. Collecting terms, their explicit form is

$$F_\tau(\theta) = g_0(\theta) + \sum_{i=1}^{20} a_i(\theta) \min_{k \in \{X,Y,Z\}} v_{i,k}(\theta), \quad (3)$$

where the branch values $v_{i,k}$ and the remainder g_0 are smooth wherever the maximum-entropy terms are. The coefficients $a_i \geq 0$ are as follows: a root weight for the six root sites, one for the twelve interior sites and the level-two site, and τ for the matrix-size site.

The workflow in Figure 1 connects numerical search with exact checking; §4 describes the individual steps.

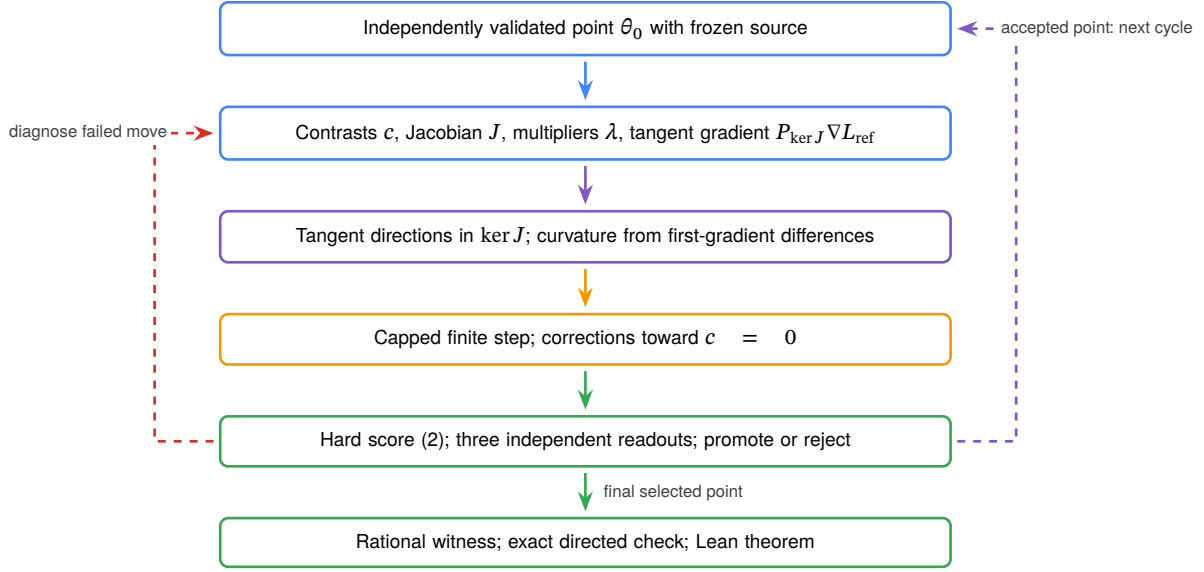


Figure 1: The search cycle. Solid arrows are the path of a candidate; dashed arrows are how finite outcomes shape the next model. Only the final frozen witness enters the certificate check.

3. Local branch geometry

At the best points we measured, the twenty sites approach three-way ties. At a model-construction point θ_0 , choose a minimum-attaining reference branch $k_i^\star$ at each site. The forty normalized *contrasts* are

$$c_{i,k}(\theta) = \frac{a_i(\theta)(v_{i,k}(\theta) - v_{i,k_i^\star}(\theta))}{M(\theta_0)}, \quad k \neq k_i^\star, \quad (4)$$

collected into $c(\theta) \in \mathbb{R}^{40}$ with Jacobian $J = Dc(\theta_0) \in \mathbb{R}^{40 \times n}$. The *tie manifold* is $\mathcal{T} = \{\theta \in \mathbb{R}^n : c(\theta) = 0\}$. At any $\theta \in \mathcal{T}$ where $Dc(\theta)$ has full row rank, $\mathcal{T}$ is locally a smooth submanifold of codimension forty, with tangent space $\ker Dc(\theta)$ and normal space range $Dc(\theta)^\top$. At a measured construction point θ_0 near $\mathcal{T}$, we use $J = Dc(\theta_0)$ for the local tangent approximation.

3.1 Measurements at the selected numerical point

We measured point H, the selected numerical point with score 2.371174031813689, using its frozen certification source, CPU float64, and the original active-coordinate masks. One forward evaluation and forty contrast derivatives plus the reference gradient reproduced the saved score, branch values, root weights and mass arrays without changing the point. The diagnostic uses the search target $\tau = 2.371170$ and the point's fixed normalization $M_0 = 6.053106932142217$ in (4) and (7).

Ties. Across all twenty sites, the largest spread $\max_k v_{i,k} - \min_k v_{i,k}$ is approximately 5.09×10^{-10} (Figure 2a). All twenty sites are three-way ties within both 10^{-7} and 10^{-8} ; all therefore also have at least two branches within each tolerance. Nineteen sites contribute to retained exponents and the twentieth to matrix size, as detailed in Appendix A.

Rank and positivity. The forty contrast gradients have numerical rank forty. The row-normalized Gram eigenvalues are approximately 0.0238861 to 3.542652, using a relative eigenvalue cutoff of 10^{-10} times the largest eigenvalue. The sixty fitted convex weights lie in $[0.32978, 0.33892]$ and are all strictly positive (Figure 2b).

Tangent remainder. At the same point, $\|P_{\ker J} \nabla L_{\text{ref}}\| \simeq 1.557 \times 10^{-6}$ and $\|\nabla L_{\text{ref}}\| \simeq 9.691 \times 10^{-2}$, a ratio of approximately 1.607×10^{-5} . The reference gradient is predominantly normal to the approximate tie manifold, with a small nonzero tangent remainder.

Branch identities. Between the previous model origin, point G, and point H, the minimum-attaining branch changes at 15 of the twenty sites. These are floating-point argmin labels among nearly tied values, rather than persistent branch

identities. The geometric quantities above concern the numerical point; exact acceptance of its rationalized witness is a separate check (§6).

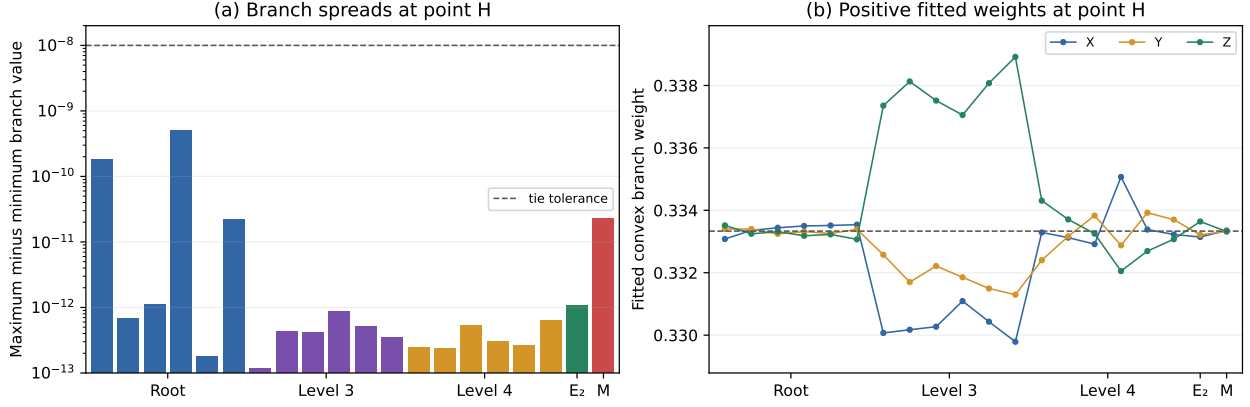


Figure 2: Branch geometry measured at the selected numerical point H. (a) Per-site branch spreads; all twenty sites meet both tie tolerances. (b) Fitted convex weights from the forty-contrast projection. Both panels use the same point and frozen source. The accompanying figure data give every branch value and fitted weight.

3.2 Exact mixture and branch slack

For each site choose convex weights $w_i \in \Delta_3$ and define the branch slack

$$s_i(\theta) = \sum_k w_{i,k} v_{i,k}(\theta) - \min_k v_{i,k}(\theta) \geq 0. \quad (5)$$

Replacing each minimum in (3) by its weighted average gives a smooth *mixture surplus* F_{mix} , and

$$F_\tau = F_{\text{mix}} - S, \quad S = \sum_i a_i s_i \geq 0, \quad \text{so} \quad \Delta F_\tau = \Delta F_{\text{mix}} - \Delta S \quad (6)$$

for any two points, even when the weights and coefficients change between them. This is exact algebra, not a Taylor model. It is also elementary: at a tie, the Clarke subdifferential of a minimum of smooth functions is the convex hull of the active gradients (Clarke 1983), and (6) is the finite version of that fact. Its use is diagnostic. A move can have positive mixture gain and still lose if slack grows faster. On $\mathcal{J}$ every s_i vanishes, so $F_\tau = F_{\text{mix}}$ there *for every choice of weights*.

We proved (6), the vanishing of slack at exact ties, and conditional finite-gain corollaries in Lean for the real scalar model. These research lemmas guide the search. They are not premises of the final theorem and are not part of the released certificate artifact.

The multipliers enter through a normalized loss. Hold the reference branches and weights fixed during each local model, write $\lambda \in \mathbb{R}^{40}$ for the non-reference weights, and put $M_0 = M(\theta_0)$. Since $F_{\text{mix}} = F_{\text{ref}} + M_0 \lambda^\top c$, define

$$L_{\text{ref}}(\theta) = -F_{\text{ref}}(\theta)/M_0, \quad L_\lambda(\theta) = L_{\text{ref}}(\theta) - \lambda^\top c(\theta) = -F_{\text{mix}}(\theta)/M_0, \quad \nabla L_\lambda(\theta_0) = \nabla L_{\text{ref}}(\theta_0) - J^\top \lambda. \quad (7)$$

Here F_{ref} uses the reference branch at every site. Solving $\min_\lambda \|\nabla L_{\text{ref}} - J^\top \lambda\|$ selects the unique weights, when J has full row rank, for which the mixture-loss gradient lies in $\ker J$. The resulting vector is $P_{\ker J} \nabla L_{\text{ref}}$, the natural first-order stationarity measure on the manifold. If the implied weights, λ together with $1 - \sum \lambda$ per site, are all nonnegative, this vector is a Clarke subgradient of the normalized hard loss at an exact tie. Its norm then measures distance from Clarke stationarity.

Equation (6) isolates the finite penalty for departing from the tie manifold. The identity itself is exact for the real scalar model; the measured values of its terms are floating-point observations.

3.3 Geometric interpretation

Write the problem as the minimization of $-F_\tau$. Each term $-a_i \min_k v_{i,k} = a_i \max_k (-v_{i,k})$ is then a maximum of smooth functions. A finite sum of such terms, with linearly independent differences of active gradients, is the standard example of a function that is partly smooth relative to the set where the active branches tie (Lewis 2002). Along that set the function is smooth. Across it the function is sharp, growing at first order in every normal direction. Partial smoothness is paired with a nondegeneracy condition: zero lies in the *relative interior* of the subdifferential, so the multipliers are strictly positive. The measured weights are strictly interior to the simplex. Full numerical rank, positive fitted multipliers and a small tangent gradient at the measured points are consistent with this interpretation; the exact hypotheses remain unproved.

Three consequences shaped the search.

1. **Surrogate gain is the wrong acceptance test.** A move with a normal component leaves $\mathcal{T}$, where F_τ is sharp and F_{mix} is smooth. By (6), such a move pays branch slack at first order in the normal displacement. The failed finite move in §5.1 demonstrates this penalty.
2. **Branch identity is not a useful state variable.** Methods that differentiate through a minimum receive the gradient of one active branch. Near $\mathcal{T}$ that choice changes from step to step, which is the classic zig-zag of first-order methods on max-type functions (Burke, Lewis, and Overton 2005). The manifold, not the active branch, is the stable object.
3. **The right local model lives in $\ker J$.** Second-order information is useful along the manifold, where the objective is smooth. Across it, first-order restoration of $c = 0$ is enough. This is the $\mathcal{V}\mathcal{U}$ decomposition (Lemaréchal, Oustry, and Sagastizábal 2000; Mifflin and Sagastizábal 2005), with $\mathcal{U} = \ker J$ and $\mathcal{V} = \text{range } J^\top$.

The problem is nonconvex, so the superlinear convergence theory for convex $\mathcal{V}\mathcal{U}$ methods does not transfer automatically.

4. Branch-aware search method

The workflow in §2, Figure 1, includes feedback from finite trials: a failed move changes the next local model. The exact checker receives only a selected, frozen point.

4.1 Fresh geometry and tangent projection

Each main cycle begins at an independently validated point θ_0 . The search recomputes the branch values, the forty contrasts and their gradients (one vector–Jacobian product each), and a reference gradient. It checks the rank of J and the sign conditions on the implied weights, and forms the tangent gradient $P_{\ker J} \nabla L_{\text{ref}}$ by rank-checked solves with the 40×40 Gram matrix. Every derivative is bound to the point at which it was computed.

Derivatives are refreshed rather than transported. At one earlier point (score 2.3711914), a twelve-direction basis carried over from previous cycles captured only 4.59% of the squared norm of the fresh tangent gradient. The omitted 95.4% motivated expanding the basis, but it did not by itself predict a large finite gain. Stale geometry is still useful as a preconditioner (§4.2), but it is never described as a fresh derivative at a later point.

4.2 Reduced tangent models

The method builds m directions $U = [u_1, \dots, u_m] \in \mathbb{R}^{n \times m}$ with $u_j \in \ker J$ to numerical tolerance; m ranges from 6 to 24 in the earlier campaign (defined in §5); the subsequent continuation used widths 6, 12 and 20 in its completed comparisons (§5.2). The directions come from the tangent gradient, residuals of earlier reduced models, and retained directions of previous cycles. Holding the reference branches and λ from (7) fixed, it fits the normalized mixture loss L_λ on their span:

$$q(z) = \gamma^\top z + \frac{1}{2} z^\top H z, \quad \gamma = U^\top \nabla L_\lambda(\theta_0), \quad He_j \approx U^\top \frac{\nabla L_\lambda(\theta_0 + hu_j) - \nabla L_\lambda(\theta_0 - hu_j)}{2h}, \quad (8)$$

where e_j is the j th coordinate vector in $\mathbb{R}^m$. Curvature thus comes from central differences of fresh first gradients, a standard matrix-free technique (Nocedal and Wright 2006). Each column is computed at two step sizes h . The reduced matrix is symmetrized and checked for agreement between the two scales, scalar-slope consistency, mixed-product symmetry, and positive definiteness before a stationary displacement is proposed. Agreement between two scales

checks the consistency of a numerical proposal. Reduced Hessians are poorly conditioned. One model in the earlier campaign had eigenvalues from 2.25×10^{-6} to 1.44×10^{-3} .

Curvature is estimated from first-gradient differences because the native maximum-entropy backward pass did not support correct second derivatives; §4.5 gives the proof-based test that confirmed this defect.

4.3 Finite steps and corrections

The positive-definite reduced loss model proposes $z^\star = -H^{-1}\gamma$, with modeled normalized mixture gain $-q(z^\star)$ and a cap on the maximum absolute raw-coordinate displacement. In the earlier campaign’s final step, that displacement was 0.679 against a cap of 0.70, so it was not rescaled. The raw-coordinate step $Uz^\star$ is evaluated at several bounded fractions, each mapped to a physical point by Φ . At each fraction, the search applies a bounded number of Gauss–Newton corrections toward the manifold (up to three in the earlier campaign’s final step and up to five in the later continuation),

$$\delta = -J^\top(JJ^\top)^{-1}(c(\theta) - c^{\text{target}}), \quad (9)$$

with domain checks on every represented parameter. The choice of c^{target} is informed by the slack identity (§4.5). Early experiments restored the contrasts inherited from the origin, $c^{\text{target}} = c(\theta_0)$. That preserved whatever slack the origin carried. Later experiments used $c^{\text{target}} = 0$ wherever the implied weights were valid convex weights. A first-order correction does not guarantee that the endpoint is balanced or better, so every corrected endpoint is re-evaluated.

All corrected and uncorrected endpoints compete on the hard score (2). Model gain, residual contrast, and slack are recorded as explanations, not as selection criteria.

4.4 Promotion

A candidate that improves the hard score is read three times:

- by the production evaluator, with 400 maximum-entropy iterations;
- by an independently implemented reference evaluator, with 1,600 iterations;
- by the production evaluator again, after serialization and reload.

An independent process then reconstructs the move, the score components, site values, root weights, masses, and the selection. Only then is the point a *validated numerical incumbent*. The final selected point passed all three readouts. The selected point is then rationalized and frozen for exact certification (§6). No gradient, curvature estimate, or research lemma enters the final theorem.

4.5 Proof as a design instrument

Lean supplied test oracles and made the conditions of proposed search steps explicit.

A proved zero-penalty identity supplied a curvature test oracle. On 54 split domains whose feasible law is fixed by its marginals, the maximum-entropy penalty is identically zero, so its second derivative along any path must vanish. On one such domain, double backward returned $0.7213 = 1/(2 \ln 2)$, while a central second difference returned 2.2×10^{-11} . The native solver’s custom backward returned the correct first derivative but treated it as a constant, silently omitting maximum-entropy curvature on the second pass. This test confirmed the missing-curvature defect first identified by code inspection of the custom backward pass, which used a saved gradient detached from its input, and motivated using finite differences of fresh first gradients. A correct direct Hessian–vector product (Pearlmutter 1994) would require implicit second differentiation through the solver’s optimality conditions (Eisenberger et al. 2022).

The elementary identity (6) organized acceptance and diagnosis. Writing slack as a separate term made failed moves interpretable and kept endpoint selection focused on the hard score rather than model gain. The identity is an accounting tool, rather than a new optimization theorem.

Zero slack at ties changed the correction target. Exactly balanced branches have zero slack. A conditional lemma bounds the hard gain of a finite move from below,

$$\frac{\Delta F_\tau}{M_0} \geq \underbrace{\eta(1 - \frac{K\eta}{2})\|v\|^2}_{\text{modeled mixture gain}} - \underbrace{B_{\text{corr}}}_{\text{correction cost}} - \underbrace{B_{\text{slack}}}_{\text{slack change}}, \quad (10)$$

where v is the tangent direction and η the step. K , B_{corr} , and B_{slack} are explicit premises: a curvature bound, a correction bound, and a slack bound. Sampled gradients suggest values for them but do not prove them. What (10) made visible is that restoring *inherited* contrasts keeps B_{slack} at the origin’s value, whereas targeting zero contrast can make it negative. The zero-contrast experiment in §5.1 tested that distinction.

Orthogonality clarified basis expansion. If a represented gradient component and an omitted one are orthogonal, the omitted norm does not reduce the represented slope. Omitted energy is therefore a reason to add directions, not a penalty on the direction already chosen.

These lemmas identify which quantity to measure next and which target to correct toward.

5. Experimental evidence

The experiments comprise an earlier campaign and subsequent continuation. Here, the *earlier campaign* means the bounded eight-hour branch-aware search from the zero-contrast correction point to numerical score 2.371176825455252; the subsequent continuation is reported in §5.2. Each expensive trial began at an independently validated checkpoint and recorded its source, origin, prediction, finite-trial controls and hard-score readout. Both phases ran on one Apple M1 Max laptop with 64 GiB unified memory, on CPU in float64 with one numerical worker thread. No GPU or cloud compute was used. Recorded optimization runs total approximately 483 process-elapsed hours, a partial sum that excludes later derivative, model-building and certification work (Appendix B).

Dupont et al. report approximately five hours on a single GPU per candidate optimization-program evaluation in their AlphaEvolve search (Dupont et al. 2026, sec. 3.3). Our eight-hour earlier campaign covers multiple model steps, corrections and unsuccessful trials, rather than one program evaluation. Appendix B records that campaign and the available partial optimizer costs; §6 records exact certification costs separately. The result demonstrates that refining and machine-checking this level-four bound is feasible on a consumer laptop.

5.1 Three tests that changed the search

A surrogate gain paid for by slack. In the finite-slack experiment, an uncorrected finite move gained 1.34×10^{-6} in normalized mixture surplus, but branch slack rose by 2.95×10^{-6} . The hard surplus therefore fell, as the exact identity (6) requires. Three finite corrections preserved essentially all the mixture gain while removing the excess slack. This falsified mixture gain as an acceptance criterion for these moves.

Correcting inherited slack. The zero-contrast experiment changed the target from the origin’s contrasts to zero where the implied weights were valid convex weights. Its validated score improved by 2.61×10^{-7} , almost entirely through slack removal; mixture gain was only 1.1×10^{-11} . The conditional Lean lemma (10) suggested the test.

A reduced model that predicted a finite gain. The model expanded an inherited six-direction basis to twenty-four directions. It used 73 full forward evaluations, 72 first-order vector–Jacobian products, and no double backward; peak memory was 26.3 GB. Against a predicted gain of 1.4868×10^{-6} from its construction point, the measured gain was 1.3990×10^{-6} , 94% of the prediction. This 40-minute stage produced the earlier campaign’s endpoint.

5.2 Trajectory and model-width choices

Appendix B gives the earlier trajectory leading into the late sequence below.

From the earlier campaign’s starting point to its numerical endpoint, the validated score fell by about 1.26×10^{-5} . New tangent directions were needed after slack cleanup; repeated correction alone did not account for the later gains. The original objective, not the reduced model, selected every promoted point.

The subsequent accepted sequence is shown below. A restoration can serve as a model-construction seed without becoming the global incumbent; accepted scores must therefore be distinguished from improvement relative to a seed.

Table 1: Later independently validated numerical incumbents. These floating-point scores are distinct from the exact certified claim in §6.

Accepted point	Numerical Ω_{hard}	Recorded source of improvement
point A	2.3711754347806533	Fresh tangent model
point B	2.3711754123857607	Fresh model from a point restored closer to ties
point C	2.3711753608722477	Fresh tangent model
point D	2.3711747196635273	Matched-width model and corrections
point E	2.3711747159935173	Current-Jacobian balance restoration
point F	2.3711741392432755	Matched twelve/twenty-direction model
point G	2.3711740797278047	Twelve-direction continuation
point H	2.371174031813689	Twelve-direction continuation

The decrease from point A to point H is approximately 1.403×10^{-6} . About 86.8% occurred in the point C–D and point E–F transitions. At the point E origin, twelve directions reached 2.37117423209488; twenty directions reached 2.3711741392432755. The extra directions helped numerically, but their conservative marginal gain per recorded computation hour fell below the preset minimum gain per hour. Subsequent cycles used twelve directions.

5.3 Recorded optimizer controls

In an earlier comparison from the same starting point, Adam beat a first-order bundle/headroom corrector with nearly equal runtimes. That corrector did not use the reduced second-order tangent models of §4.2.

Table 2: Both methods used the same starting point and nearly equal runtimes; Adam’s listed score was last measured at 4,365 s.

Method	Common origin Ω_{hard}	Recorded runtime (s)	Last measured hard score
Bundle/headroom corrector	2.3713403708334013	4546.10	2.371338993431329
Fresh Adam	2.3713403708334013	4551.90	2.3713326744345458

A later comparison started Adam and a search using fresh projected gradients with balance corrections from the same point (score 2.3711899235254914), allowing 1,100 s each; both stopped before completing their planned steps. The last saved corrected score was 2.3711897212798734; Adam’s was 2.3712622190785293, worse than the starting point. These partial results cannot be compared at exactly equal elapsed times.

A static block preconditioner improved its origin but lost to the same-origin fresh-model control, and two structured perturbations lost to unperturbed continuation. These mixed controls motivate reporting the methods as an adaptive case study. Appendix B collects the compute costs.

6. Exact certification of the selected point

The selected numerical point has score 2.371174031813689. Its physical parameters were rationalized with 64-bit precision, and fresh maximum-entropy witnesses were constructed. The canonical certificate has **458,921,565 bytes** and SHA-256

dfcede72c9dbf8eaf7b55f34cdd56d93dca7fe8a28fca2787466bec410f1873f

A directed exact checker verifies the domain conditions, maximum-entropy enclosures and the sufficient feasibility inequality (1) for

$$\Omega = \frac{43740440563687076465}{2^{64}} = 2.37117403423113 \dots$$

$$\frac{2371177}{10^6} - \Omega = \frac{854824676828237063}{288230376151711744 \cdot 10^6} \approx 2.96577 \times 10^{-6}.$$

The difference between the certified claim and the printed numerical score is approximately 2.41744×10^{-9} . It includes rationalization and outward certification bounds; it is not an estimate of floating-point error or a measurement of rationalization alone.

The generated module is `MatrixMath.Generated.Omega_dfcede72c9dbf8ea`. Its acceptance theorem checks the complete literal certificate bytes, claim and digest. Its exponent theorem, `omegaResult_dfcede72c9dbf8eaf...`, states $\omega_{\mathbb{Q}} \leq \Omega$ for the definitionally specified rational-field exponent. The complete declaration names, elaborated statements and their hashes are supplied in `compiled-assurance.json` (Appendix C).

Trust boundary. Profile CN evaluates the closed acceptance computation natively. Its compiled theorem depends on one certificate-specific `native_decide` axiom. The exponent theorem additionally depends on `MatrixMath.AX1_combination_loss`, which supplies the mathematical bridge from the checked feasibility conditions to the bound on ω . That bridge implements the cited combination-loss theorem (Alman et al. 2025) but is not proved within the released Lean package. Both statements also use `Classical.choice`, `propext` and `Quot.sound`. CN trusts the relevant pinned compiler, native runtime and big-integer implementation; Lean kernel soundness is a metatheoretic assumption. This is not an axiom-free or kernel-only proof. A separately implemented exact Rust checker accepts the same certificate as an independent cross-check, not as a replacement for the Lean theorem.

Two full CN executions completed on the Apple M1 Max laptop with Lean 4.33.0 and GMP 6.3.0: the initial proof and a new evaluation after retrieval from an initially empty local content-addressed store. Their complete `prove` commands took 14,468.57 s and 14,521.79 s (approximately four hours each), including the Rust prerequisite, generation, dependency work and compiled auditing. Standalone Rust checks took 813.53 s initially and 825.72 s after retrieval. These are command durations, not isolated Lean kernel times or complete research costs. Strict compiled-statement and axiom audits, complete trust ledgers and four deterministic artifact-byte comparisons passed. The second computation was a new whole-module execution, rather than reuse of the first acceptance result.

Appendix C gives the matching source identities and reader instructions. Removing the project axiom and replacing native evaluation with kernel-checkable full-certificate acceptance are ongoing work; neither strengthening is claimed here.

7. Implications and limitations

The exact check establishes a feasible level-four point; the exponent bound follows from Alman et al.’s published combination-loss theorem, represented by the unproved Lean bridge disclosed in §6. The search case study connects near-tie geometry with practical tangent models and finite corrections; proof-based tests supplied exact accounting and confirmed a derivative defect. The published level-three and level-four values differ by about 1.6×10^{-4} , so further numerical refinement operates on a much smaller scale. Dupont et al. suggest that larger improvements likely require new mathematical ideas. Known barriers for analyses based on the Coppersmith–Winograd tensor exclude $\omega = 2$ by such methods (Ambainis, Filmus, and Le Gall 2015; Alman and Vassilevska Williams 2023; Christandl, Vrana, and Zuiddam 2021).

Limitations. This is one adaptive search trajectory, with local controls rather than a systematic comparison of optimizers. Geometry was measured at a few points in one basin; full rank, positivity and partial smoothness have not been proved for the exact point. Curvature was sampled, not bounded, and the conditional finite-gain lemmas were not instantiated on the winning trajectory. The certificate validates the selected endpoint, not its search history. No local or global optimality, convergence rate, or general optimizer superiority is claimed.

Open questions.

1. *Which local geometry transfers beyond this basin?* The branch-weight investigation in Appendix E found no construction-specific explanation or demonstrated gain from imposing equal weights.
2. *Can the tie manifold be parameterized directly?* Forty equality constraints on 4.66 million coordinates remove little dimension, but an explicit chart of $\mathcal{T}$ would turn the problem into smooth optimization and might make level five tractable.

3. *Can local optimality be certified?* On a partly smooth, nondegenerate point, second-order conditions on $\ker J$ give a local optimality criterion. An exact version would bound what further local search at this level can achieve. It would not bound global search.

Appendix

A. Level-four construction and notation

Fix the Coppersmith–Winograd parameter $q = 5$ and recursion level $\ell^* = 4$. A level- ℓ shape is a nonnegative integer triple with coordinate sum 2^ℓ . The root chooses a region and a shape. Positive internal nodes choose regions and admissible splits, zero-shape nodes carry split distributions, and positive level-two leaves carry one scalar parameter. Distinct paths remain distinct even when their levels, shapes, and regions coincide. The complete statement is that of (Alman et al. 2025), in the form given by (Dupont et al. 2026). Our mathematical specification transcribes it clause by clause, and the certificate checker of §6 decides it exactly.

The twenty minimum sites in (3) have the following accounting:

Table 3: Minimum sites in the level-four objective.

Site class	Count	Coefficient a_i
Root regions	6	Corresponding root weight
Level-three regions	6	1
Level-four regions	6	1
Level-two exponent E_2	1	1
Matrix-size exponent M	1	Target τ

The physical point p consists of distributions and leaf scalars. The raw search coordinates θ are mapped to $p = \Phi(\theta)$ by softmax and sigmoid maps, with inactive coordinates fixed. The score (2) is only defined by the displayed quotient when $M(\theta) > 0$; exact domain checks are part of certification.

B. Earlier trajectory, record mapping and compute costs

The late point A–H sequence appears once, in §5.2. Earlier selected points are listed here.

Table 4: Selected validated numerical points. Floating-point scores; the certified bound is in §6.

Stage	Validated Ω_{hard}	What the step tested
Lift of level-three point	2.372151067	First level-four point, symmetric parameterization
Distinct-path parameter search	2.3713528479	Per-path untied parameters
Continued distinct-path Adam search	2.3712878268	Continued from the distinct-path point
Twelve-direction model	2.3711914229	Twelve-direction tangent model with corrections
Fresh omitted-direction step	2.3711912647	Finite step in a freshly measured omitted direction
Finite-slack experiment	2.3711899235	Six fresh tangent directions
Zero-contrast correction	2.3711894607	Corrections toward zero contrast
Campaign, first cycle	2.3711879899	Fresh tangent model after zero-contrast correction
Campaign, model union	2.3711877592	Union of same-origin models
Campaign, fresh geometry	2.3711859220	New incumbent after refreshed geometry
Campaign, basis expansion	2.3711826240	Expanded tangent basis
Campaign, retained metric	2.3711777689	Six-direction parent of the final model

Stage	Validated Ω_{hard}	What the step tested
Capped twenty-four-direction step	2.3711768255	24-direction model, three corrections

Compute costs. The earlier campaign defined in §5 took 8 h 08 min 05 s, including 1 h 31 min 02 s of unsuccessful stages and repairs. Its method-family accounting is:

Table 5: Aggregate costs by campaign stage, rounded to minutes. The remaining elapsed time includes planning, reporting and gaps between stages. This timing ends before the late point A–H continuation.

Method family	Elapsed, including failures	Recorded improvement
Fresh tangent models and same-origin model unions	2 h 21 min	7.31×10^{-6}
Static block preconditioning	18 min	0
Structured perturbations with unperturbed control	23 min	7.7×10^{-8} , all from control
Preconditioned continuation, recovery and extensions	4 h 34 min	5.25×10^{-6}

The available optimizer records contain approximately 483 process-elapsed hours; the union of their recorded start/end windows covers about 277 hours. These are partial process durations and overlapping-window coverage, not CPU time or end-to-end research cost. Later derivative, model and certification stages use separate records and are excluded. Whole-research elapsed and CPU totals remain unknown. Section 6 gives the selected certificate’s checking durations separately.

Mapping to the research record. These identifiers are retained solely for locating the original files; the paper uses descriptive labels.

Paper label	Internal record identifier
Point A	best008; fresh tangent cycle 006
Point B	best009; fresh tangent cycle 007
Point C	best010; fresh tangent cycle 008
Point D	best011; matched-width cycle 009
Point E	best012; current-Jacobian restoration
Point F	best013; width comparison cycle 010
Point G	best014; narrow continuation cycle 011
Point H, selected numerical point	best015; narrow continuation cycle 012
Distinct-path parameter search	P575
Continued distinct-path Adam search	P1950 label; symmetric-basin campaign control at update 2000
Twelve-direction model	0141
Fresh omitted-direction step	0151
Finite-slack experiment	0154
Zero-contrast correction	0157
Capped twenty-four-direction step	0181

C. Selected package and reader verification

The matching package is `certifications/l4-2026-10-01-best015`. Its `verification-runbook.md` and `trust-boundary.md` are the authoritative reader instructions for this certificate.

The result identities are:

- Certificate: 458,921,565 canonical bytes, with the SHA-256 in §6.
- Generated Lean module: 458,924,665 raw bytes. Its raw SHA-256 is:

```
71d36d830605c27c77ae116ba37a091241db4feb9a1ce746bf4772160cf289a2
```

Compressed payload: 0mega_dfcede72c9dbf8ea.lean.gz.

- Original frozen normative source manifest: 1,243 files, SHA-256:

```
668e1f1c6f0bfdbfabe840b6162cda4efebe79d59b0c528f966ab3b0e9ef6fee
```

- Original Lean source closure recorded in assurance.json:

```
0b7add6f623b0254565d2f18bfe9d29253b16c3467153c3d3de62073ecccef66
```

The reduced reader workspace is distinct from those original full-source identities. Its inventory identifies unchanged checker/proof sources and package-specific build wrappers. Its Cargo lock matches the included workspace and retains the recorded dependency versions and checksums; `take-manifest.json` fixes the Lean dependencies. A reader's new source closure and runtime trust ledger must be recorded separately.

1. **Check identity.** With `certificate.json` and the compressed theorem in `payloads/`, run `python3 verify-files.py` in the certification directory. It checks sizes, hashes and lossless theorem transport. A successful identity check is not mathematical acceptance.
2. **Build in an isolated copy.** Follow the runbook using Rust 1.94.0, Lean 4.33.0 and the supplied dependency locks. Build the Rust CLI and Lean checker/soundness modules. Do not regenerate dependency locks to bypass a failure. Building these modules does not evaluate the selected certificate.
3. **Run the independent exact check.** From the copied verification-source directory, run `target/release/mm verify ../payloads/certificate.json --skip-lean --json`. Require a successful exit, the matching hash and exact rational claim, and `rust_cross_check: ok`.
4. **Run full CN certification.** Run `target/release/mm prove ../payloads/certificate.json --profile cn --json`. Require successful full evaluation and compiled auditing. Compare the regenerated module's raw hash with the inventory, and compare compiled declaration kinds, statements, statement digests and transitive axioms with the supplied assurance. Review the new execution's trust ledger.

D. AI assistance

AI systems were used substantially for engineering, analysis, proof development and LaTeX writing, under the author's direction. This assistance is distinct from the numerical optimization loop: AI systems helped design, implement and analyze experiments, while search programs evaluated and ranked candidates using the hard objective and independent checks. The agents' outputs were treated as proposals and passed the same recorded checks as any other search artifact. No claim depends on who, or what, wrote the search code: the bound rests on the certificate and the checker under the disclosed trust assumptions. A reconstruction of the working instructions is given in the supplementary material.

E. Near-uniform weights: an inconclusive investigation

A study of stored branch-gradient systems examined the near-uniform fitted weights using alternative linear solvers, consistent reference-branch rebasing and moderate block metrics. The pattern persisted under those controls, but its cause remained unexplained. Fitted Euclidean projection weights can depend on the coordinate metric away from exact stationarity; forcing equal weights left additional normal-gradient residual. A positive, full-rank stationary synthetic example with unequal weights refuted a universal explanation based on ties and positivity alone. Global tensor symmetry does not force equal weights at asymmetric points. The study found no construction-specific theorem, no justified reduction to equal weights, and no demonstrated improvement from imposing them.

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